THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** electricity	Electric energy.
2	**(d)** copper	Good conductor.
3	**(a)** cap	Metal cap.
4	**(c)** two or more cells	Combination of cells.
5	**(d)** plastic	Insulation.
6	**(d)** flat base	Flat.
7	**(a)** filament	Thin wire.
8	**(a)** safety	Protects circuits.
9	**(d)** connections using symbols	Standard drawing.
10	**(a)** open or close a circuit	On/off.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A source of electricity (1); It has chemicals (1).	1+1
12	Copper (1); Iron (1).	1+1
13	A broken path (1); No current (1).	1+1
14	Combination of cells (1); Gives more voltage (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell (1); Bulb (1); Switch (1).	1+1+1
16	Cell: long and short lines (1); Bulb: circle with a cross (1); Switch: a gap with a lever (1).	1+1+1
17	To turn on (1); To turn off (1); To save energy (1).	1+1+1
	OR: Copper (1); Iron (1); Aluminium (1).	1+1+1
18	Connect a cell (1); Close the switch (1); Use wires (1).	1+1+1
	OR: Copper wire (1); Iron nail (1); Graphite (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take a cell, a bulb and wires (1); Join one end of the bulb to + of the cell (1); Join the other end to - of the cell (1); The circuit is complete (1); The bulb glows (1).	1+1+1+1+1
	OR: Light (1); Heat (1); Fan (1); TV (1); Cooking (1).	1+1+1+1+1
20	Copper (1); Iron (1); Aluminium (1); Rubber (1); Plastic (1).	1+1+1+1+1
	OR: Too much current (1); Wire heats (1); Melts (1); Circuit breaks (1); Appliance safe (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** shock	Dangerous.
2	**(c)** open	No path.
3	**(d)** does not flow	No path.
4	**(b)** heating effect of current	Wire heats.
5	**(a)** carefully	Safety.
6	**(d)** flows	Complete.
7	**(a)** heating effect of current	Heating.
8	**(c)** when they are not in use	Save energy.
9	**(b)** allows electricity to pass	E.g. metal.
10	**(d)** the cell through the circuit	Along the wires.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rubber (1); Plastic (1).	1+1
12	A complete path (1); Current flows (1).	1+1
13	A complete path for current (1); Includes cell and bulb (1).	1+1
14	Thin wire in a bulb (1); It glows (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rubber (1); Plastic (1); Wood (1).	1+1+1
16	Circuit is broken (1); No current (1); Filament is cold (1).	1+1+1
17	Eraser (1); Plastic comb (1); Wooden spoon (1).	1+1+1
	OR: Do not touch wires (1); Use dry hands (1); Use a fuse (1).	1+1+1
18	To protect from heavy current (1); It breaks the circuit (1); Prevents fires (1).	1+1+1
	OR: Carries more current (1); Does not heat (1); Safe (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dry hands (1); Do not overload (1); No damaged wires (1); Use a fuse (1); Switch off (1).	1+1+1+1+1
	OR: LED uses less power (1); Lasts longer (1); Cooler (1); Cheaper to run (1); Better for environment (1).	1+1+1+1+1
20	Has a heating coil (1); Current flows (1); Coil gets hot (1); Heat passes to the plate (1); Clothes are pressed (1).	1+1+1+1+1
	OR: Take three cells (1); Connect + to - (1); Join with wires (1); Voltage adds (1); Brighter bulb (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
- Confusing series and parallel: in series there is *one* path, in parallel there are *many* paths.
- Thinking all materials with a plastic covering are conductors; the covering is an **insulator**, the wire inside is the conductor.
- Writing that the fuse is connected in parallel with the appliance; it is connected in **series**.
- Mixing up the heating effect (heat from current) with the cause of the glow of a bulb; the filament glows because it becomes very hot.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** on	Closed.
2	**(d)** does not allow electricity to pass	E.g. rubber.
3	**(b)** tungsten	High melting point.
4	**(c)** metal	Metal.
5	**(d)** copper wire	Metal.
6	**(a)** light emitting diode	Energy saving.
7	**(d)** energy	Electric energy.
8	**(c)** rubber	Does not conduct.
9	**(c)** LED bulb	LED.
10	**(b)** safety switch	Automatic.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is an insulator (1); Prevents shocks (1).	1+1
12	Opens and closes a circuit (1); Controls the bulb (1).	1+1
13	A safety wire (1); Melts on high current (1).	1+1
14	Heating (1); Used in an iron (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron (1); Heater (1); Toaster (1).	1+1+1
16	Iron (1); Geyser (1); Heater (1).	1+1+1
17	Fuse (1); MCB (1); Insulating gloves (1).	1+1+1
	OR: Light (1); Fan (1); Fridge (1).	1+1+1
18	TV (1); Fan (1); Washing machine (1).	1+1+1
	OR: Water conducts (1); Shock risk (1); Short circuit (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Open: broken (1); No current (1); Closed: complete (1); Current flows (1); Bulb glows (1).	1+1+1+1+1
	OR: Never touch sockets (1); Do not play with wires (1); Never use wet hands (1); Ask adults (1); Keep away from kites near power lines (1).	1+1+1+1+1
20	Glass (1); Filament (1); Wires (1); Metal cap (1); Gas (1).	1+1+1+1+1
	OR: Conductor lets current pass (1); Insulator does not (1); Copper (1); Rubber (1); Use of both (1).	1+1+1+1+1

Mistakes to avoid:

- Connecting a cell with the wrong polarity or leaving a gap in the circuit and still expecting the bulb to glow; it needs a **closed** circuit.
- Saying that a fuse wire has a high melting point; it has a **low** melting point so that it melts first.
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